

Cave: A Computational Framework for Subjective Trajectories

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1 I. The Problem

Plato's cave is often introduced as a lesson about knowledge: prisoners mistake shadows for reality, philosophy turns the soul toward the source, and truth is recovered by escaping the chamber of appearances. We retain that intuition, but we shift its use. For our purposes, the cave is not primarily a moral image of ignorance, nor a metaphor for illusion. It is a design constraint. If appearances are partial, delayed, filtered, and organized from a particular position, then a model of experience cannot begin by assuming direct access to the world-as-given.

Cave therefore treats the allegory as an engineering problem. We ask what must be represented if we want to model not merely an external sequence of events, but the way such a sequence is available to a situated subject. The answer cannot be a simple input log. It must include sensors, limits, memory, expectation, error, attention, value, and the consequences of prior state. In this sense, the cave is not a place from which the model must escape. It is the condition under which the model becomes meaningful: a controlled setting in which relations between world, signal, and subject can be made explicit.

The first methodological step, then, is to name the object we are trying to build. We do not begin with a theory of inner life and then search for a picture that might illustrate it. We begin with an episode in which something happens outside a subject, and we ask how that episode becomes available from the subject's position. The relevant object is therefore not the external sequence itself, nor an image projected from it, but the ordered record of how the subject is changed by being coupled to it.

This record must be relational. At each moment, the subject has some prior state; only some aspects of the environment are available; only some of what is available is taken up; and what is taken up alters what can be expected, remembered, valued, or ignored later. Thus the modeled trajectory is not a container of private facts. It is a path through state space under constraints: a history of selections, omissions, anticipations, mismatches, and updates. By making that path explicit, we give the problem a computational target: the carried-forward structure of subject-world coupling through time.

We therefore treat the question "what is consciousness?" as premature for the present purpose. Our narrower question is which relations must persist across time for an episode to become a subject-side trajectory rather than a replay of external inputs. A passive copy can list what occurred. It cannot say what was available, what was missed, what was expected, what mattered, or how one moment altered the next. For that, the model must carry relations such as sensed $\rightarrow$ retained, retained $\rightarrow$ expected, expected $\rightarrow$ actual, error $\rightarrow$ learning, and state $\rightarrow$ future exposure.

This shift is deliberately modest, but it is also decisive. We do not need to settle the nature of consciousness in order to specify the computational

burden of situated appearance. If two subjects receive the same external sequence but differ in sensorium, attention, memory, value, or prior expectation, then the relevant modeled object cannot be the sequence alone. It must be the evolving relation between that sequence and each subject's state. Cave begins there: with the minimal machinery required for an input history to acquire a perspective-dependent path, and with the insistence that this path be explicit enough to inspect, compare, and test.

2 II. Functions As Seeds

We begin with a deliberately modest claim. If our aim is to model the path a subject takes through an episode, we need not first endow the subject with a complete private theater, a vocabulary of qualia, or a solved theory of consciousness. We can start lower, with operations that any temporally situated system must perform if the present is to matter for what follows.

The important shift is to treat these operations as subject-side relations rather than as neutral records of an external sequence. A stimulus is not merely given; it is encountered from some position, through some channel, under some limit, against some retained state. Likewise, a later state is not merely later in the world; it is later for a system whose memory, expectation, and exposure have been altered by what came before. In this sense, the primitive materials of the framework are not images or reports, but transformations.

This is why functions matter. They let us specify, before any psychological interpretation is added, what must be carried from one moment to the next. A minimal model of experience-like structure can therefore begin with small directed relations:

First, there is a positional relation, *here* $\rightarrow$ *there*: a system can distinguish its present locus of contact from some other locus toward which contact may move. Second, there is a temporal relation, *before* $\rightarrow$ *after*: the present state is not isolated, but ordered with respect to what has already occurred and what may follow. Third, there is a retention relation, *input* $\rightarrow$ *memory*: what is encountered can leave a trace rather than vanish with its presentation. Fourth, there is a projective relation, *memory* $\rightarrow$ *expectation*: retained structure can constrain what the system is prepared to receive next.

Once expectation is present, comparison becomes possible. We obtain *expected* $\rightarrow$ *actual*, not as a philosophical drama, but as an ordinary computable contrast between a prior state and a current signal. From that contrast follows *error* $\rightarrow$ *learning*: mismatch can alter the parameters by which later signals are interpreted. Finally, there is *state* $\rightarrow$ *future exposure*: the subject's updated condition can affect which parts of the world it will meet strongly, weakly, or not at all.

These are mathematical objects before they are psychological ones: maps,

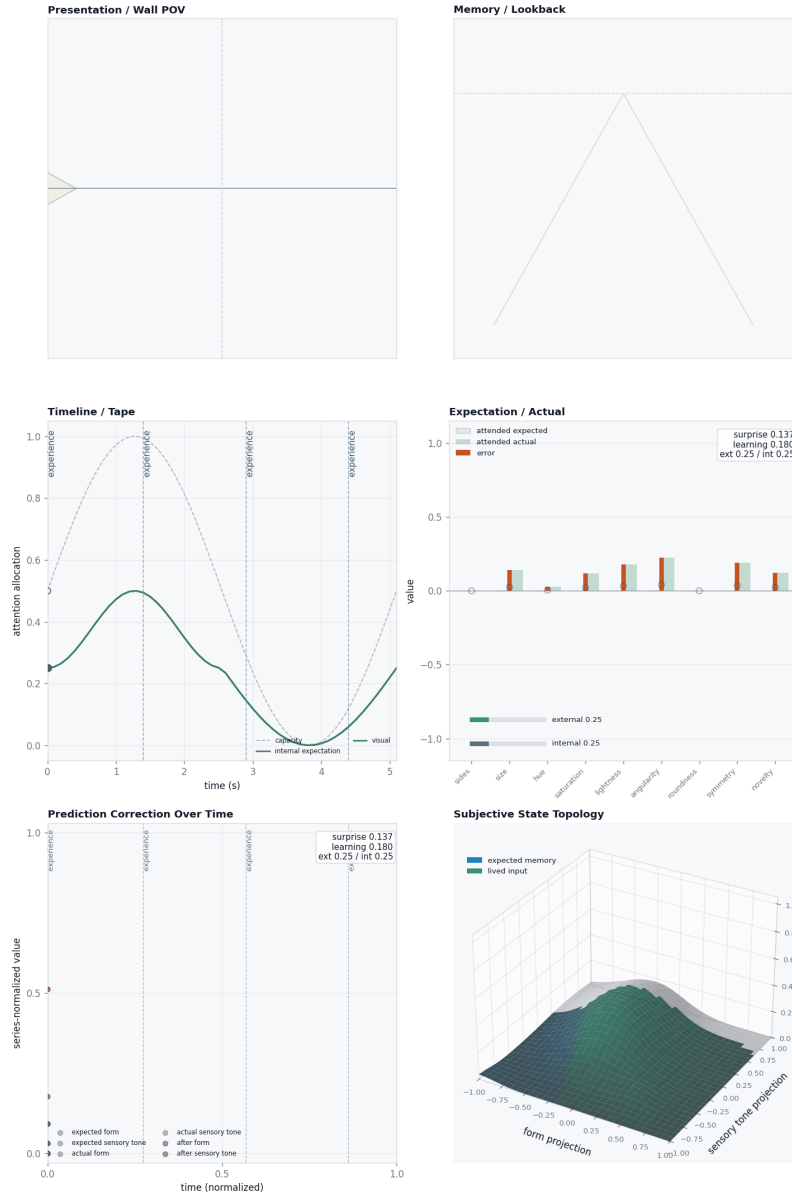


Figure 1: Multiple synchronized projections of one serialized episode. The panels are views over the same subject-side trajectory, not separate simulations.

updates, comparisons, traces, and policies.

To make such functions do the work we need, we must bind them to a particular subject. Their domain is not the world in itself, but the portion of the world that can be sensed, selected, retained, and updated by a configured system. Thus $input \rightarrow memory$ is not a universal inscription of the stimulus. It is an update conditioned by sensorium, attention, prior state, and capacity. Likewise, $memory \rightarrow expectation$ is not a prophecy about the world as such. It is a subject-side projection from what this system has carried forward.

The functions must also be history-sensitive. The same object encountered twice is not, for the subject, simply the same event repeated. The first encounter can alter memory; the altered memory changes expectation; the changed expectation changes the error produced by the second encounter. They must be limited, because no subject receives all features with equal force. And they must be consequential, because what is sensed or missed can alter later learning, value, and exposure.

Once these conditions are imposed, the model no longer produces a bare copy of an external sequence. It produces a path through that sequence as taken by a subject. That path, rather than the world alone or an image of the world, is the object Cave sets out to model.

3 III. What Is A Subjective Trajectory

By a *subjective trajectory* we mean the ordered succession of states on the side of the modeled subject. An episode may contain many external objects, but the trajectory is not simply their chronological list. At each timestep, the subject has a current state: what its sensors can transduce, what attention admits, what memory already contains, what it expects, what errors it registers, what it values, and how the present step changes what later steps can become.

This definition lets us separate the external course of events from the subject's carried history. Two subjects may encounter the same episode while producing different trajectories, because their sensors, priors, attentional capacities, memories, or exposure policies differ. Conversely, the same subject may produce a different trajectory on a second encounter, because the first encounter has altered memory and expectation.

Thus the trajectory is a temporally extended computational object. It is built from timestep updates, but it is not reducible to any single frame. Its significance lies in the continuity across frames: expectation before input, error after comparison, learning after error, memory after attention, and future exposure after state change. This is the object whose status we now delimit.

We should therefore be precise about the status of the object. A subjective trajectory is not introduced as an assertion that the model has found,

captured, or explained consciousness. It is a disciplined intermediate object: the record of how an episode becomes subject-side under specified mechanisms. The claim is not metaphysical first; it is architectural. If we specify what the world presents, what the subject can sense, how it attends, predicts, errs, learns, remembers, values, and modulates exposure, then the resulting path is no longer merely the external sequence.

In compact form, the construction is

$$\begin{aligned} &\text{external sequence} + \text{subject profile} + \text{timestep state updates} \\ &\longrightarrow \text{episode observations} \longrightarrow \text{subjective trajectory.} \end{aligned}$$

This formula is meant as a modeling commitment. The external sequence supplies temporally extended objects. The subject profile supplies the configured sensorium, priors, memory, attention, value, and action constraints. The update rule supplies the recurrence by which prior state conditions present registration and present registration changes later state. Episode observations are the inspectable trace of those updates. The subjective trajectory is the ordered whole obtained from them.

Thus we do not begin by asking whether this object is consciousness. We begin by making explicit the relations that any later theory would have to discuss.

For this reason, every downstream artifact should be treated as an instrument of inspection, not as a rival object. A wall view may display what was active and attended; a dashboard may expose surprise, learning rate, or value; an embedding may place whole runs near or far; a topology surface may accumulate the weighted trace of contact; an animation may make change perceptible to us. None of these adds a second subject behind the first. They are different coordinate systems laid over the same serialized path.

The distinction is methodological but consequential. If a renderer, comparison metric, or explanatory diagram is allowed to create its own state, then we have lost the thing we meant to study: the relation carried forward through time by one modeled subject. The rendering may select, compress, color, smooth, align, or compare; it may help us notice structure that is difficult to see in raw observations. But it must remain downstream of the trajectory. A disagreement between two views should therefore be read first as a difference in projection, scale, or summary, not as evidence that the episode contains multiple hidden experiential objects. The ontology of the framework stays deliberately sparse: external objects, configured subject, update rule, observations, trajectory, and projections that read it.

4 IV. Experience Objects

We begin the model outside the subject. An *experience object* is the smallest unit of external presentation that the framework treats as available for possible uptake. Formally, we may think of an object o_i as carrying an interval $I_i = [t_i^{\text{on}}, t_i^{\text{off}})$, a feature description, a modality label, a salience value, and parameters governing how strongly it can contribute to learning if it is sensed. It may also carry presentation metadata, but that metadata is auxiliary: it helps a later view display or name the object; it does not by itself determine the subject’s state.

This definition is intentionally modest. We do not begin by asking whether the object is seen, remembered, understood, liked, or consciously noticed. Those are downstream questions, answered only after the object encounters a configured subject with particular sensors, attentional limits, priors, and memory. At this level, the object is simply a temporally available source of structure. Its duration matters because the subject may couple to it weakly, strongly, intermittently, or not at all across its active interval. Thus the external unit is not an instantaneous datum but a supported opportunity for subject-side transformation.

We therefore distinguish an object’s support from its later display. A wall position, a pixel coordinate, a glyph in a memory corridor, or a point on a topology surface is a way of reading a run after the model has already produced subject-side states. It is not the object itself. The object is the temporally extended carrier of features that may or may not become available to a subject through its sensorium.

This distinction lets very different cases share one formal status. A triangle may have visual features over an interval; a tone may have auditory features; a word or token may have linguistic features; a conversational turn may carry semantic and social structure. None of these begins as a command to draw something at (x, y) . Each begins as something externally present over time, capable of entering a trajectory only through the subject’s configured channels and update rules.

Thus, when we write an object o_i , we are not smuggling in a viewer. We are specifying an opportunity for coupling. If the relevant sensor is absent, the object remains externally active but internally silent. If attention is weak, its contribution is correspondingly diminished. The model therefore preserves a clean order: world-side availability first, subject-side uptake second, rendering last.

For practical authoring, however, it is useful to allow documents to speak in a richer descriptive vocabulary. An input file may say that an object is large, novel, beautiful, threatening, or painful. We treat such terms as an authoring layer, not as a second ontology of experience. Before the run, they are resolved into the same fields already available to the model: feature magnitude, salience, modality-specific intensity, value parameters, learning

weight, or metadata. The update rule then sees only the ordinary object description.

This matters because many intuitive qualities are not identical to the state variables they may influence. A painful object is not automatically surprising; an expected pain remains painful. A beautiful object is not automatically attended; it may receive higher value or salience, but attention is still allocated by the subject profile and state. Likewise, authored novelty should not bypass the prediction mechanism. It may mark the object as unusual in the world description, but whether it is unexpected for this subject depends on prior memory and prediction.

The aim is therefore ergonomic rather than metaphysical. Authors may describe episodes in human terms, while the model preserves a single computational path. Qualities are compiled into object fields; subject-side consequences arise only through sensing, attention, expectation, value, learning, and memory.

5 V. Subject Profiles

We call a subject profile the part of the model that is fixed before an episode begins. It specifies what kinds of signals can be transduced, what feature vocabulary those signals inhabit, how attention is allocated, how much information enters the workspace, how expectations are formed, how errors change future expectations, and how traces are retained. It also specifies value sensitivities, modest action or exposure policies, topology accumulation rules, and the initial state from which the first prediction is made.

This is easiest to understand as the experimental preparation of a subject. Before showing any object, we must know whether the subject can hear, see, parse words, distinguish colors, remember recent inputs, assign negative value to threat, or devote attention to an internal expectation channel. These choices are not annotations on a later picture. They determine the update functions that will operate at each timestep.

Formally, a profile supplies the parameters and initial conditions for the transition rule that maps prior state and currently available objects into the next subject-side state. It is therefore the place where limits, priors, capacities, sensitivities, and policies enter the run. Without such a profile, an episode is merely an external ordering of inputs; with one, it becomes a candidate trajectory for a particular modeled subject.

Consequently, the profile is not a label attached to an otherwise fixed run. It is an argument of the run itself. If we write the external episode as $x_{1:T}$ and the profile as p , then the modeled trajectory is better understood as $\tau(p, x_{1:T})$ than as a property of $x_{1:T}$ alone. Changing p changes the transition rule, the available channels, the gain on those channels, the prediction made before the next input arrives, and the traces that survive after the object is

gone.

This is why subject-dependence enters before visualization. A renderer may later choose a wall, graph, surface, or table, but all of these read from a trajectory that has already been shaped by sensors, attention, memory, value, and policy. The important variation is therefore not that two pictures of the same state look different. It is that the state being projected can differ because the modeled subject was differently prepared to receive, select, predict, and retain the episode.

The profile also prevents a common ambiguity. We do not ask only what was presented. We ask what could be transduced, what was attended, what was expected, what counted as error, and what was allowed to modify future state.

This distinction gives the profile a direct diagnostic role. If an active object carries an auditory feature and the profile contains no audio transducer, the object remains part of the external episode, but it does not enter the subject-side input. In notation, it may belong to O_t while contributing nothing to O'_t . It can be present for an omniscient observer of the run, and it can still be listed among active inputs, but it is absent from the modeled subject's sensed state. No later rendering should reintroduce it as if it had been heard.

A second diagnostic concerns priors. Two subjects may receive the same transduced input with the same immediate input effect, yet differ in observed memory because their memory states did not begin alike. One subject may already expect a pattern, while another treats it as novel. The update caused by the presented object can be identical relative to each subject's baseline, while the resulting memory vector differs because prior state is part of what is observed after the update.

For this reason, comparisons must say whether they measure resulting state or state change. The former includes the subject's history; the latter isolates the contribution of the current episode. Both are legitimate, but they answer different questions.

6 VI. The State Update

At a given time t , the model first determines which external objects are temporally active. This is only the availability condition; it is not yet subjective presence. Exposure then specifies how strongly the subject remains coupled to those active objects. The sensing stage applies the subject's sensorium, so a modality for which no sensor exists can remain present in the world while making no contribution to the internal state. Attention next gates both the externally sensed channels and the internally predicted channels. In this way, capacity and allocation determine not merely what is displayed, but what can enter subsequent computation.

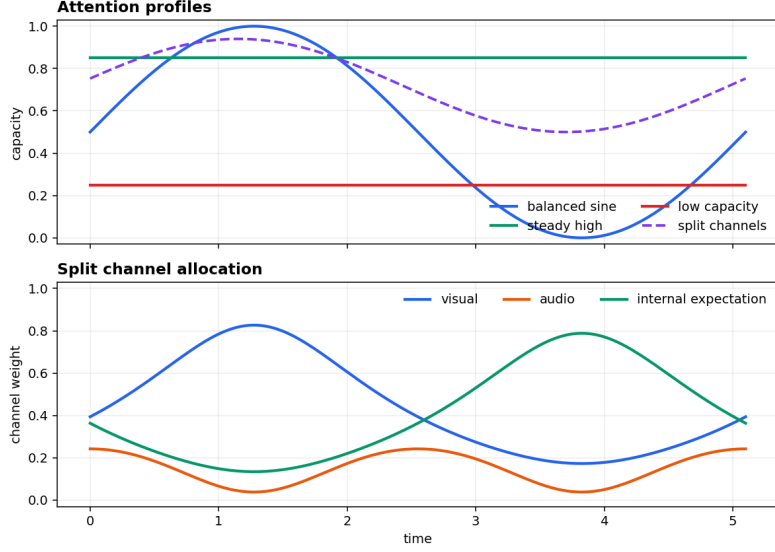


Figure 2: Subject profiles differ in admission capacity and channel allocation before any renderer is involved. These curves are parameters of the subject-side update path, not overlays added after the run.

The gated input is compressed into a workspace representation suited to comparison and update. Before memory is changed, the current prediction is read from prior state and compared with the actual attended input. Their difference yields prediction error, from which surprise and the effective learning rate are derived. Only then is vector memory updated, and only from the subject-side signal that survived sensing and attention. Value and objective terms are evaluated separately from prediction error, preserving the distinction between mismatch and desirability. Finally, the event deposits into topology according to its attended and learned effect, giving the episode a cumulative geometry.

We therefore treat the update not as a metaphorical flow, but as a composition of maps. Given a prior subject state, the episode supplies only the currently available world-side material; the model then asks how that material is transformed under the subject’s profile and current condition. In schematic form, one may write

$$S_{t+1} = F_t(S_t, O_t; \theta),$$

where O_t denotes the active external material and θ denotes the configured subject. The important point is that F_t is not a single opaque operation. It is a structured composition of smaller functions, each of which reads and writes subject-side quantities.

This gives the trajectory its status as a computational object. A trajectory is not the input sequence itself, nor a later visualization of that sequence.

It is the ordered result of applying these state transformations through time:

$$S_0, S_1, \dots, S_T.$$

If the subject profile changes, the same O_t can yield a different path. If the prior state changes, the same stimulus can be predicted, missed, amplified, ignored, valued, or learned differently. Thus the episode becomes experience-like only after it has passed through the subject-dependent update chain.

For this reason, we store more than the final visible result of a step. The full scene state S_t records the intermediate products by which the step was produced:

$$S_t = (t, O_t, O'_t, \text{sensors}, A_t, U_t, W_t, E_t, P_t, \sigma_t, \eta_t, M_t, V_t, J_t, a_t, L_t).$$

Here O_t is availability, O'_t is exposed or sensed material, A_t attention, U_t the attended subject-side input, W_t workspace content, E_t expectation, P_t prediction error, σ_t surprise, η_t learning rate, M_t memory, V_t value, J_t objectives, a_t action or exposure modulation, and L_t accumulated topology. The notation is not meant to imply that every implementation must expose exactly these names. It states the architectural requirement: the run must preserve the quantities from which later interpretations are read.

Consequently, a report, plot, wall rendering, memory view, or subject comparison is a projection from S_t , or from the ordered sequence of such states. It may select, aggregate, or re-encode state, but it may not become the state it depicts. This prevents a change in visualization from silently becoming a change in model dynamics. No view owns S_t .

7 VII. Attention And The Gating Rule

Attention is not a curve to be drawn over an otherwise complete trajectory. It is one of the mechanisms by which the trajectory is made. We therefore represent attention at time t by two coupled quantities: a scalar capacity c_t , which determines how much attentional force is available, and a channel distribution α_t , which determines where that force is allocated. The distribution may range over visual, auditory, linguistic, affective, proprioceptive, or other modeled channels, with $\sum_i \alpha_{t,i} = 1$ when capacity is nonzero.

This separation is important. A subject may be highly attentive but narrowly allocated, or broadly allocated but weakly engaged. The model should distinguish both cases. Capacity answers the question, “how open is the subject to state change at this moment?” Distribution answers the question, “through which routes may that change occur?” Together they determine an attention effect, which we may write schematically as $\psi(c_t)\alpha_t$.

The gate is not merely descriptive. It controls how strongly available signals enter sensing, comparison, learning, memory, and later topological

accumulation. Attention is therefore not an annotation on experience; it is a condition on whether and how candidate inputs become subject-side consequences.

For each external modality m , the available object-side signal is first limited by the subject’s sensorium and then scaled by the corresponding attentional allocation. Schematically, if $x_{t,m}$ is the sensed signal made available by active objects, the subject-side contribution is not $x_{t,m}$ itself, but

$$\tilde{x}_{t,m} = \psi(c_t)\alpha_{t,m}x_{t,m}.$$

The same rule applies to the internal channel. The expectation used at time t is a prior-state prediction, computed before current input is allowed to update memory. But this prediction also must be admitted into the workspace. Thus the model treats expected input as a signal with its own attentional cost:

$$\tilde{e}_t = \psi(c_t)\alpha_{t,\text{exp}}e_t.$$

Prediction is therefore not an invisible background possession of the subject. It is a readout from prior state, gated like any other contributor to the present update.

This symmetry is important because error is computed between admitted actual input and admitted expected input. A subject may fail to register an object, fail to consult an expectation, or partially do both. In the limiting case $c_t = 0$, we require $\psi(c_t) = 0$. External presence may still be recorded as availability, but no subject-side sensing, comparison, learning, memory increment, or topological deposit follows from it.

The resulting rule can be stated sharply: attention changes gain, not coordinates. If an object has a feature representation $\phi(o)$ in the modeled vocabulary, the attentional gate does not rewrite $\phi(o)$, move it to a different region of feature space, or turn one modality into another. It changes the strength with which that representation is allowed to contribute to S_t . The same holds for expectation. To attend to an expectation is not to invent a new prediction, but to admit a prior-state prediction into comparison, error, learning, and memory with some effective weight.

This distinction prevents a common confusion. A dimly attended object is not a different object in feature geometry; it is an object with reduced subject-side consequence. Conversely, a highly attended object does not become more semantically central by definition. It becomes more causally effective in the update. The difference appears in sensing strength, prediction error, learning rate, memory increment, value evaluation, and topology deposit, not in the object’s basic feature address.

We treat this as an executable property. Holding objects, features, sensors, and priors fixed, changes in c_t or α_t should alter state effects without altering the underlying feature embedding. In the limiting test, $c_t = 0$ collapses the relevant update contributions. Attention is therefore a gate in the model, not a metaphor in the prose.

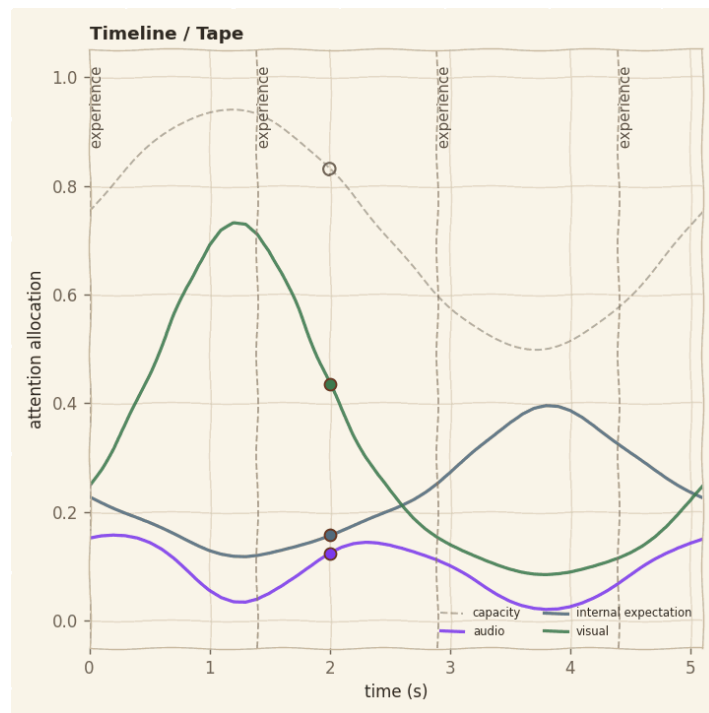


Figure 3: Temporal support and attention allocation define when external availability becomes effective uptake. The timeline view reads object intervals and attention values from the episode; it does not create them.

8 VIII. Expectation, Error, And Learning

We begin with a simple constraint: the subject must predict from the state it already has, not from the state it is about to acquire. At time t , the relevant prior is M_{t-1} , together with the current attentional allocation, sensor profile, and other subject-side variables. From that prior state the model forms an expected input, $\hat{x}_t$. Only then is the currently sensed input, x_t , compared against it. The difference $x_t - \hat{x}_t$ is the prediction error available to learning.

This is not merely an implementation preference. It is the condition that makes expectation a genuine temporal relation. If the present input were allowed to update memory before the expectation were computed, the model would in effect consult the answer before making the prediction. The expected state would already contain the event it is supposed to anticipate, and error would be artificially suppressed.

Thus the update rule must preserve a before-and-after structure: prior memory supports expectation; expectation is gated and exposed; actual input arrives through the subject's sensors; comparison produces error and surprise; learning then modifies memory. We treat this ordering as part of the model's meaning, not as a display convention.

We can see the point most clearly in the ordinary case of repetition. Suppose an object o appears across several successive intervals with sufficiently stable features. On the first encounter, the subject has little basis for anticipating it. After several attended encounters, however, the recurrent state has been shaped by that history. The model should therefore begin to assign o -like features to the next expected input. Repetition has become expectation, not because the external world has changed, but because the subject-side history now carries the regularity.

Now consider a later interval in which the regularity fails: the object is absent, displaced, altered, or replaced by an incompatible feature pattern. In that case the discrepancy is not merely another item in the stream. It is a difference between what the subject was prepared to receive and what the sensors actually deliver. The violation appears as prediction error, and its magnitude contributes to surprise and to the subsequent learning update.

This is why the temporal ordering matters for even the simplest demonstrations. If the present input is folded into memory before expectation is evaluated, then both repetition and violation are partially absorbed before they can be measured. The system can still record change, but it has lost the relation that makes novelty intelligible: an actual present measured against a prior expectation.

A further consequence is that expectation is not a passive backdrop. In the update chain, the predicted vector is available only insofar as the subject allocates capacity to the internal expectation channel. Let α_t^{exp} denote that allocation, and let $\psi(c_t)$ denote the current attentional effect. Then $\hat{x}_t$ may be present as a computable prediction, but its influence on comparison is

mediated by this gate. With low allocation, the prior may contain relevant structure while exerting little force on the moment. With high allocation, the prior becomes readable, and mismatch can become correspondingly sharper.

This distinction matters because we do not want prediction to function as an omnipresent oracle. A subject may have learned a regularity and yet fail to bring it to bear under distraction, overload, or competing salience. Error is therefore not simply the distance between present sensation and everything memory could imply. It is the distance between actual input and the portion of prior state currently exposed for comparison.

By making expectation spend attention, the model treats anticipation as a subject-side act rather than a background assumption. A run may fail to show surprise not because the world was predictable, but because the subject did not attend to its own prediction. That failure is itself part of the trajectory.

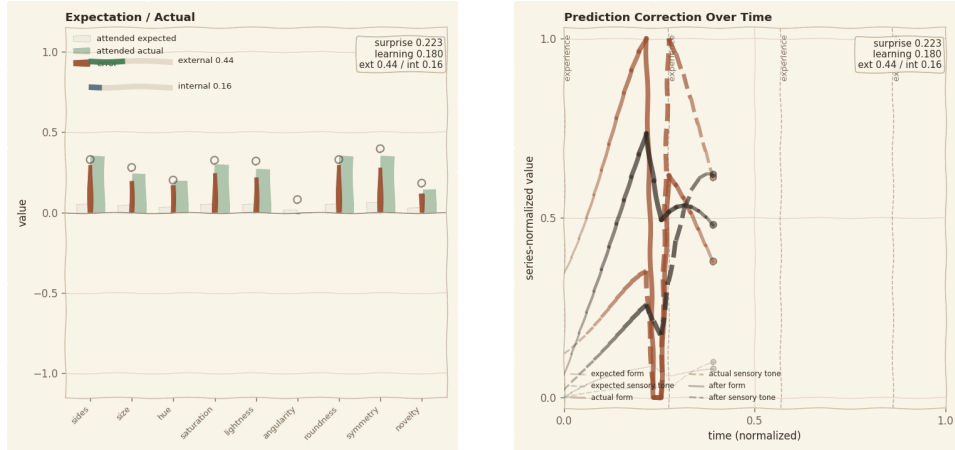


Figure 4: Expectation is read from prior state, compared with admitted input, and corrected through learning. The correction panel is diagnostic; it does not update the model.

9 IX. Memory

We distinguish memory into two coupled but non-identical registers. The first is vector memory: a recurrent numerical state carried forward by the subject and used to form expectations, trajectory embeddings, and comparisons across runs. It is the memory most directly involved in prediction. It need not resemble a list of remembered objects; rather, it is a compressed trace of what has affected the subject so far.

The second is object memory. This register concerns temporally extended inputs as retained traces: a shape that was present, a word that was heard,

a tone that persisted, an event that occupied an interval. Object memory supports lookback, inspection, and object-level reports. It asks not only what the current predictive state has become, but which encountered things have left identifiable traces and with what remaining strength.

These two forms should not be collapsed. A subject may acquire a predictive bias without retaining a clear object trace, or may retain an object-level trace whose predictive contribution has decayed. Conversely, strong attention to an object may strengthen both registers, but through different mechanisms. The distinction lets us separate memory as recurrent state from memory as retained episode structure, while keeping both inside the same subjective trajectory.

For vector memory, the decisive point is that the update is not a direct inscription of O_t , the externally active object set. It is an inscription of what the subject has actually transduced and admitted into state. Formally, if x_t denotes the modality-specific features made available by the world, the memory update depends on a subject-side signal such as

$$u_t = \psi(c_t) \alpha_t \odot S(x_t),$$

where S represents the subject’s sensors, α_t the channel allocation, and $\psi(c_t)$ the scalar attention effect. The recurrent memory is then updated from u_t , not from x_t alone:

$$M_{t+1} = F(M_t, u_t, \eta_t).$$

This small distinction carries much of the model’s subject dependence. An object may be present in the episode, and therefore available to an external observer of the sequence, while making no contribution to vector memory for a particular subject. If the relevant sensor is absent, $S(x_t) = 0$ for that modality. If attention is fully withdrawn, $\psi(c_t) = 0$, and the effective input is again null. In both cases the object is not merely dimly rendered; it fails to enter the recurrent update. Later expectations, embeddings, and comparisons should therefore reflect what was carried through the subject, not what was merely present outside it.

For object memory, the natural unit is the completed interval. While an input o is active, from t_{on} to t_{off} , the model accumulates the degree to which that object has affected the subject. Only when the interval closes do we finalize a retained trace. Its initial strength may be written schematically as

$$r_o^0 = \sum_{t=t_{\text{on}}}^{t_{\text{off}}} \psi(c_t) \alpha_t(o) q_o(t),$$

where $q_o(t)$ denotes the transduced and relevant object contribution at that moment. This makes the retained trace depend on exposure, sensing, and attention across the object’s duration, not on its mere external existence.

After finalization, the trace is carried as an object-level memory with age-dependent decay, for example

$$r_o(\tau) = r_o^0 D(\tau),$$

with $D(\tau)$ a decreasing retention function. Thus a recently and strongly attended object remains available for lookback, while an unattended, unsensed, or long-past object may leave little or no recoverable trace.

The lookback corridor is therefore not an independent memory mechanism. It is a projection of these finalized traces, ordered and displayed for inspection. It reads object memory; it does not create it, update vector memory, or alter the trajectory it renders.

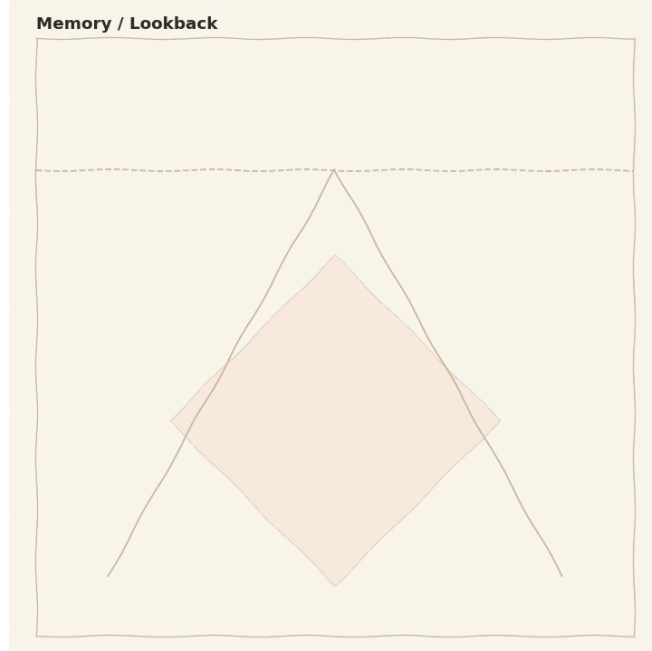


Figure 5: Memory lookback projects retained traces after object intervals close. The corridor is a view of object memory state, not the storage format itself.

10 X. Value, Objective State, And Exposure

We begin with a small but important refusal: we do not identify mismatch with badness. The comparison between expected and actual input tells us how well the current signal fits the subject's prior state. It does not tell us whether the signal is wanted, harmful, useful, or aversive. A painful stimulus may be perfectly expected; a pleasant event may arrive as a surprise.

Conversely, an accurate prediction of threat remains threat, and a prediction error around relief may be beneficial rather than damaging.

For this reason, Cave treats prediction and valuation as different update channels. Prediction error is computed from the relation between expected and actual subject-side input. Value is computed from evaluative mappings over sensed features, object identity, bodily or task variables, and the subject’s configured aims. The two channels may interact downstream: high surprise can increase learning, and value can change future exposure. But neither channel is reduced to the other. This separation prevents a common mistake in experience models, where salience, novelty, pain, desire, and error collapse into one scalar. We instead require the trajectory to carry a structured account of what was unexpected and a distinct account of what mattered.

We implement this distinction by giving the subject two evaluative layers. The first is a set of valence evaluators. A valence evaluator reads the currently sensed, attended object features, together with any retained object identity or task label, and assigns signed or categorical marks: pleasant, painful, useful, costly, safe, threatening, and so on. These marks are not required to agree with prediction. They are attributes of how the subject is configured to care about the input. Thus an object may receive low prediction error and high negative valence, or high prediction error and positive valence.

The second layer is an objective evaluator. Here we collect several costs and benefits into a utility-like state variable, J_t , or a vector of such variables when the subject has multiple aims. A simple evaluator may combine prediction cost, attention cost, workspace or compression cost, pain penalties, pleasure rewards, and task progress. The important point is not the particular moral theory or reward function chosen, but the location of the computation. Objective state is computed after sensing and comparison, from subject-side quantities already present in S_t . It is therefore inspectable, replaceable, and separable from surprise. Once computed, it can bias what the subject permits to affect it next.

Exposure is the point at which these evaluative signals become consequential for the subsequent run. Let ρ_t denote a coupling policy over the active objects and channels at time t . The policy does not invent new objects, rewrite the episode, or plan an arbitrary future. It changes how strongly already active or soon-active inputs are permitted to reach the subject: whether the subject dwells, averts, attenuates, samples narrowly, or remains open. In the update chain, this means that valuation can influence later exposure before later sensing, so that J_t , valence marks, and recent surprise may help determine ρ_{t+1} .

This is a deliberately modest form of agency. We do not need a full theory of action in order to model a subject whose future contact with the world is partly regulated by its own state. A subject that repeatedly attenuates

a painful but predictable object will accumulate a different memory trace from one that continues to attend to it. A subject that increases exposure to a surprising but useful signal will learn a different path than one that suppresses novelty. The external sequence may be identical in both cases, but the effective sequence is not. Exposure therefore closes a minimal loop: value changes coupling, coupling changes sensed input, and sensed input changes the trajectory that follows.

11 XI. Workspace Compression

We introduce workspace compression to make explicit that the subject does not carry the entire attended field forward. Even after sensing and attention, current input may be too large, too detailed, or too heterogeneous to become a usable state. The workspace therefore acts as a limited-capacity code for the present moment: it receives the subject-side attended input and produces a smaller representation W_t that downstream mechanisms can compare, retain, value, and learn from.

The important point is that compression is not merely storage reduction. It is a model commitment about availability. Features that survive the bottleneck are available to expectation, error, memory, and later comparison; features that do not survive may have been externally present, and even weakly sensed, without becoming effective parts of the continuing trajectory. In this sense, the workspace combines selection and information loss in one operation.

We should distinguish this from attention. Attention controls the strength and distribution of entry into the subject-side stream. Workspace compression determines what form that entered material takes once it must be represented under capacity limits. A high-attention input can still be compressed into a coarse code; a low-attention input may fail to contribute enough structure to be retained. The compressor is therefore one of the places where a trajectory becomes genuinely path-dependent: what is preserved now constrains what can be predicted, remembered, and surprised by later.

We may write the operation abstractly as

$$W_t = \kappa_\theta(U_t; K_t),$$

where U_t is the attended subject-side input, K_t is the current workspace capacity, and κ_θ is a compression rule. The rule need not be psychologically elaborate. It may preserve the largest components, the most salient features, the features most aligned with current expectation, or the features most relevant to the subject’s value and task variables. What matters is that dominance is defined from the subject side. A feature is not kept because it is objectively important in the world, but because, under this subject profile and this state, it wins access to the limited code.

The discarded remainder should not be treated as if it never existed. It belongs to the external and possibly sensed episode, but it does not become part of W_t . It may be pooled into a coarse residual, quantized below useful resolution, or removed from the carried-forward state altogether. Thus two subjects may attend to the same object and still produce different workspace states if their compressors privilege different dimensions. One may retain shape and lose color; another may retain threat and lose geometry. The compressor therefore determines not only how much is kept, but what kind of present the subject can subsequently use.

Finally, the compression operation should itself leave a trace in the state. If U_t contains more structure than W_t can carry, the loss is not merely a hidden implementation detail. We may record a compression cost C_t , measuring distortion, discarded salience, capacity pressure, or the mismatch between what entered the workspace and what could be made available downstream. This cost is a model signal. A later view may display it, but the view does not create it.

This matters because compression cost can enter objective evaluation. A subject may tolerate lossy compression when the retained code is sufficient for its current task, but incur penalty when important distinctions are collapsed. In this way C_t can contribute to J_t alongside value, surprise, and learning consequences. The model can therefore distinguish a calm, well-compressed moment from an overloaded one even if both produce a workspace vector of the same dimensionality.

It may also affect adaptive attention. If repeated high cost indicates that the present allocation is wasting capacity, the subject can shift later attention weights, reduce exposure, privilege another modality, or reserve more capacity for expected structure. Thus compression participates in the trajectory rather than merely preparing material for visualization. It helps determine what the subject can afford to carry forward.

12 XII. Topology-Like State

By a topology-like state we mean a history-bearing density over a projected feature space. Let π denote the projection from the subject-side feature representation into an inspection space, and let $L_t(x)$ be the accumulated density at position x after timestep t . The update is not a redraw of the current scene. It is a deposit into an existing field:

$$L_{t+1}(x) = \lambda L_t(x) + D_t(x),$$

where λ controls persistence or decay, and D_t is the new contribution written by the present state.

That contribution can come from several sources. Current attended input deposits mass near $\pi(U_t)$. Object memory deposits weaker or delayed

traces according to retained object strength. Prediction deposits near $\pi(E_t)$, marking what the subject was prepared to encounter. Correction geometry deposits along, or between, expected and actual locations, marking the directed discrepancy by which the state had to adjust. Thus topology is not merely where inputs have appeared; it is where inputs, expectations, retained traces, and errors have left consequences.

The important point is accumulation. A single event may be small, but repeated attention, repeated prediction, or repeated correction can carve stable structure into L_t . The coordinates are therefore an inspection coordinate system for historical subject-side effects.

We should therefore be careful about the status of L_t . It is not a proposal that experience occurs in a literal two-dimensional mental arena, nor that the subject carries an internal map with the same geometry as the plotted surface. The dimensionality, metric, and projection π are choices made for inspection. They let us ask where, in a chosen representation, subject-side effects have accumulated.

This distinction matters because the same run may be projected in several legitimate ways. One projection may emphasize sensory similarity, another value, another prediction error, another memory strength. None of these projections replaces the state update that produced the trajectory. They are instruments for reading consequences that are already present in the serialized state.

The topology-like field is useful precisely because it gives deposits a history. It can show repeated exposure, stable expectation, neglected regions, recurrent correction, or the fading of traces under decay. But a ridge in L_t is not itself a belief, a perception, or a remembered object. It is evidence that some class of subject-side update has repeatedly left mass in that region of the inspection space. Thus the surface is a disciplined model artifact: not a metaphysics of mental space, but a way to inspect how experience-like state acquires structure over time.

We should therefore read the topology surface as a diagnostic instrument, not as the modeled episode itself. The trajectory is the ordered sequence of subject-side states: what was sensed, gated, expected, missed, retained, valued, and allowed to alter later state. The surface is a projection from that sequence into an inspection space. It can reveal accumulation, neglect, recurrence, and correction, but it cannot replace the update chain that produced those effects.

This matters especially when the surface is visually persuasive. A ridge, basin, cluster, or path may invite psychological language, but the warrant for that language remains in the serialized state. If we change the projection π , the visible surface may change while the underlying run remains fixed. Conversely, two surfaces may look similar while their trajectories differ in attention, prediction, value, or memory history. The topology readout is therefore useful only when interpreted as one view among others.

In this sense, L_t belongs on the same side of the architecture as timelines, walls, dashboards, and comparison plots. It is downstream of the episode. It helps us inspect how a subject-side history has acquired structure, but it is not the subjective trajectory. The trajectory is the modeled object; the topology surface is one disciplined way of reading it.

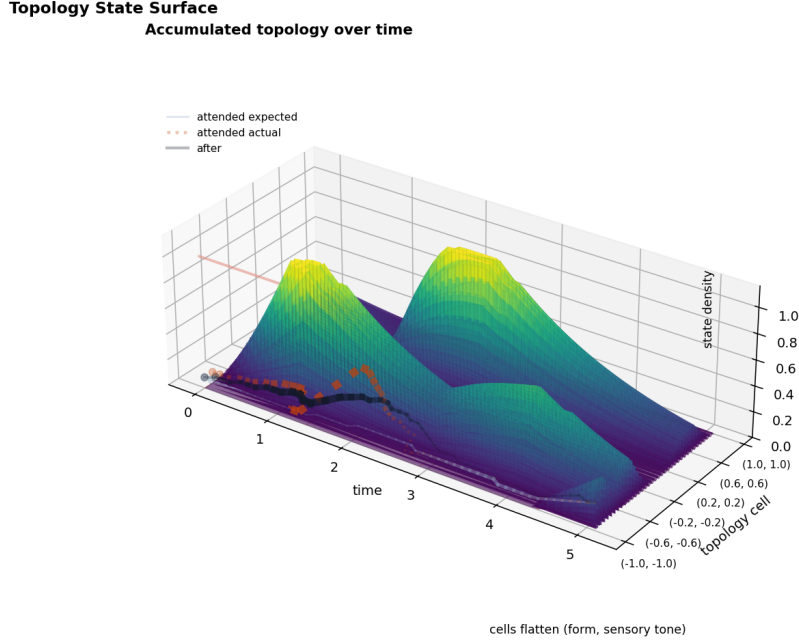


Figure 6: Topology state stacks accumulated expected and lived density through time under a chosen projection. The surface is an inspection artifact over serialized state, not literal mental geography.

13 XIII. The Architectural Rule

The first rule of the architecture is therefore deliberately austere: the run produces a state history, and every artifact shown to an observer is derived from that history. We insist on this boundary because otherwise the model can quietly split in two. One computation would determine the subject’s trajectory, while another, hidden in the visualization layer, would decide what is salient, retained, expected, or consequential. At that point the displayed scene would no longer be an inspection of the model; it would be a second model wearing the costume of a view.

In Cave, then, state has ontological priority within the formalism. The quantities updated at each timestep—sensed input, attention, expectation, error, memory, value, and accumulated topology—belong to the trajectory

itself. A view may select among these quantities, arrange them spatially, color them, aggregate them, or compare them, but it may not create new subjective facts. This restriction is not a cosmetic software preference. It is the condition under which different renderings can remain mutually accountable. If two diagrams disagree, we should be able to trace the disagreement to a projection choice, not to an unannounced recomputation of the episode. The architectural boundary therefore protects the object of study: a single subject-side state sequence.

Consequently, a renderer must be treated as an interpreter of an already completed, or currently sampled, trajectory, not as a participant in its production. When it places an object on a cave wall, draws a point along a timeline, opens a memory corridor, raises a topology surface, or lights a dashboard cell, it is choosing a coordinate system for inspection. The coordinate is not the experience object, and the graphic mark is not the subject's state. It is an address assigned to quantities that have already been computed by the update chain.

This distinction is especially important because visual forms are seductive. A wall can make the scene appear spatial, a timeline can make it appear narrative, and a surface can make it appear geometric. These appearances are useful only if we remember their derivative status. They summarize relations among active input, attention, expectation, error, memory, and value; they do not authorize new relations among them. If a visualization emphasizes a missed sound, a retained image, or a prediction failure, that emphasis must be recoverable from the serialized episode and its state fields. The view may make the trajectory legible, but it may not improve, complete, or correct it.

The practical form of this rule is a one-way rendering pipeline:

$$\text{Episode} \rightarrow \text{EpisodeFrame} \rightarrow \text{ViewState} \rightarrow \text{pixels}.$$

An Episode contains the serialized trajectory: inputs, observations, state variables, and metadata. An EpisodeFrame selects the relevant moment or interval from that trajectory. A ViewState then specifies how that selected information will be arranged for inspection—which coordinates, colors, scales, filters, or overlays will be used. Only at the final step do pixels appear. At no point does information flow backward from the displayed image into the modeled subject.

This is the architectural difference between Cave and a visualization package. A visualization package may begin with graphics and treat data as material for display. Cave begins with a subject-side computation and treats graphics as a disciplined projection of that computation. The renderer may help us notice a relation, but it cannot make that relation true. It may expose surprise, attention, memory, or value, but only if those quantities already exist in the Episode.

Thus even interactive views remain constrained. Zooming, rotating, filtering, or changing projections may alter what we see, but not what the subject underwent. The trajectory is computed once; the views are many.

14 XIV. The Episode Contract

We call the *Episode* the serialization contract because it fixes what a completed run must expose, while refusing to identify that exposure with any particular picture of the run. A renderer may later draw a wall, a corridor, a graph, or a dashboard; none of these is the canonical form of the model. The canonical form is the inspectable record from which such views are derived.

This distinction is important. If the model state lived inside the visualization, then every change of interface would risk becoming a change of theory. By contrast, an Episode lets us ask the same questions of a run regardless of how it is displayed. What did the subject predict? What arrived through the subject-side channels? What changed in memory? Where did error, attention, and learning enter the trajectory? These are model questions before they are presentation questions.

The Episode therefore serves as a boundary object between computation and interpretation. It is sufficiently structured to support comparison, debugging, and replay, but sufficiently neutral to allow different external sources to be adapted into the same inspection surface. In this sense, it is the durable trace of the subjective trajectory: not the world itself, and not a rendering of the world, but the serialized record of the subject-side path through an episode.

Concretely, an *Episode* contains a source name, a vocabulary, an array of `EpisodeInput[]` records, an array of `EpisodeObservation[]` records, a duration, and metadata. The inputs describe what was externally available during the run: their temporal support, modality, features, salience, and any authored presentation information. They do not yet say what the subject experienced. That is supplied only by the observations.

For each sampled time t , an `EpisodeObservation` records the subject-side update at that moment. It includes the expected state, the actual sensed state, the resulting memory state, surprise, learning rate, attention, attention weights, active inputs, input features, and any local metadata. These fields are deliberately redundant enough to make the run auditable. We can ask whether the expected vector was computed before the memory update, whether actual input was unavailable because no sensor transduced it, whether zero attention suppressed its effects, or whether a memory change came from prior state rather than from the presented input alone.

Thus the contract records not merely that something occurred, but how it entered the subject-side computation. The Episode is therefore a structured trace of update relations, not a transcript of external events.

Source-neutrality should not be mistaken for semantic equivalence. A native Cave sequence, a token stream, and a conversation differ in what their external objects are, how their features are authored, and what interpretive practices surround them. A triangle in a perceptual episode, a word in a text stream, and a turn in dialogue are not the same kind of thing. The claim is narrower and more useful: once adapted, each can expose the same computational questions. What was active? What could the subject sense? What was attended? What was predicted? What changed memory? What error was produced?

This shared surface is what makes the Episode a bridge to the comparative work that follows. Paper 2 does not need every source to become the same object in the world. It needs completed runs to yield comparable subject-side records. From those records we can derive memory trajectories, attended-input signatures, prediction-error paths, learning profiles, and combined subjective trajectory signatures. The Episode is therefore the hinge between modeling and comparison: it preserves the specificity of the source while normalizing the form in which the subject-side path can be inspected.

In this way, the contract completes the first construction. We have not rendered experience. We have serialized the relations a model subject carried forward through time.

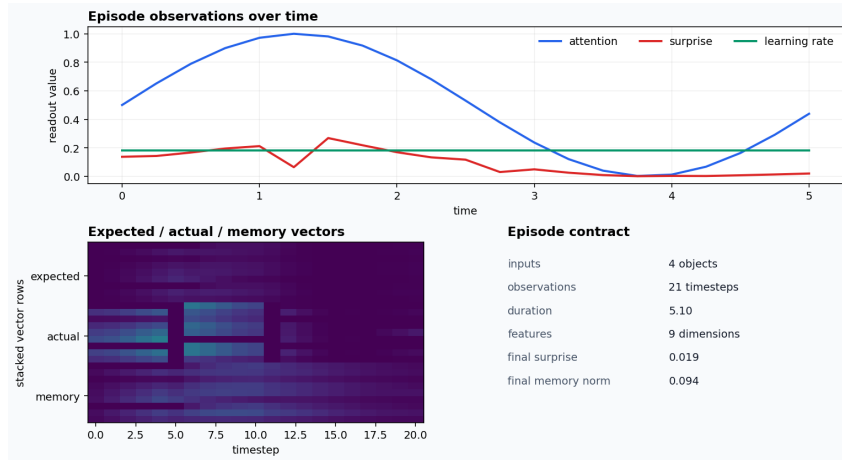


Figure 7: The episode contract exposes expected input, actual input, memory state, surprise, learning rate, attention, active inputs, and metadata as ordinary readout state.

15 XV. Views Are Projections

We treat a view as an observer of an already computed moment. At time t , the model has produced an EpisodeFrame: a compact record of active

inputs, sensed features, attention, expectation, error, memory, value, and other subject-side quantities. A view may select from this record, arrange it, scale it, color it, or animate it. But it may not revise the record it reads.

Formally, a view is a projection $v_i : F_t \rightarrow R_{i,t}$, where F_t is the frame and $R_{i,t}$ is a renderable representation. The important restriction is one-way dependence. The frame is generated from the trajectory; the view is generated from the frame. No choice of camera angle, glyph, opacity, layout, or coordinate system is allowed to write back into S_t , M_t , A_t , or any other component of model state.

This rule is deliberately simple. It lets us compare visualizations without confusing them with the computation they display. If a display needs derived quantities, those quantities belong to its ViewState, not to the underlying episode. In this sense, rendering is interpretation, not update. The trajectory remains the single source of truth.

Thus, a wall view, a timeline view, a memory-lookback view, an expectation/actual comparison, a correction diagram, and a topology surface should be understood as different questions asked of the same frame. The wall may ask which presently active objects are available to the subject and with what opacity. The timeline may ask how the episode is ordered. A lookback view may ask which completed objects have left object-memory traces. An expectation/actual display may ask how prior prediction and current sensed input differ. A correction view may ask how error is assigned under the learning rule. A topology view may ask where attention-weighted deposits accumulate across the run.

None of these views is a privileged duplicate of experience. Each suppresses some coordinates of F_t and emphasizes others. The same high-surprise moment can appear as a bright mismatch in one display, as a bend in a memory path in another, and as a local deposit on a topology surface in a third. These are not competing events. They are coordinated projections of the same state update.

We therefore impose a useful discipline: if two views disagree, we first ask whether they are reading different fields, using different scales, or applying different derived geometry. We do not infer that the subject has acquired two simultaneous trajectories.

The point can be stated as an architectural invariant. A rendered experience is not an additional experience-object and not an auxiliary subjective path. It is a reading of a path already produced by the update functions. The permitted dependence runs $\text{Episode} \rightarrow \text{EpisodeFrame} \rightarrow \text{ViewState} \rightarrow \text{pixels}$; there is no inverse dependence from pixels back into the Episode.

This is why the claim is enforceable rather than merely methodological. A renderer should receive an immutable frame, construct its own derived geometry, and return a representation. If it needs opacity, coordinates, labels, interpolants, or camera parameters, those belong to the view layer. If it changes attention, memory, expectation, value, prediction error, or expo-

sure, it has ceased to be a view and has become part of the model. That boundary can be checked in code.

The practical consequence is important. We may build many renderings of the same moment without multiplying the modeled subject. A dramatic wall animation, a sparse diagnostic dashboard, and a topology plot can all be faithful, provided they preserve one-way dependence on the same trajectory. Conversely, a beautiful rendering that silently changes S_t , M_t , or A_t is not a better visualization of experience; it is a different computation. The rendered experience is therefore a projection of the subjective trajectory, never a second trajectory.

16 XVI. Comparing Subjects

The reason comparison becomes possible is not that two subjects are assumed to share an inner world, but that their runs are written into the same inspectable form. An Episode gives us a common contract: at each time step we can ask what was active externally, what entered the subject side, what was predicted, what was missed, what changed in memory, and how strongly the event mattered to subsequent state. Once those quantities are serialized, a run is no longer merely an animation or a qualitative report. It is an ordered object over which distances, alignments, and contrasts can be defined.

This lets us separate several questions that are often conflated. We may compare two subjects exposed to the same external sequence. We may compare one subject under different priors, sensors, attention policies, or value settings. We may also compare different source domains, provided they have been adapted into the same Episode contract. In each case, the comparison is not between worlds as such, but between subject-side trajectories generated from worlds under profiles.

We should therefore treat comparison as a second-order reading of the model state. It does not create the trajectory; it measures relations among completed trajectories. The following signatures are useful ways to perform that reading.

Several embeddings are especially useful because each isolates a different relation carried by the run. The memory trajectory records how the recurrent subject state M_t moves through time. It is our most direct trace of accumulated history. The attended-input embedding records what actually entered through sensors and gates, not merely what was externally present. The expected-input embedding records the model’s prior readout P_t , allowing us to compare what the subject was prepared for with what arrived. The prediction-error embedding records the signed or vector difference between actual and expected input, while surprise σ_t and learning rate η_t summarize how consequential that mismatch became for updating.

We may also define a state-effect embedding. Here the object of interest is not the raw state, but the difference made by the episode relative to an appropriate baseline run. Finally, we can concatenate or otherwise integrate these channels into a combined subjective trajectory signature. This signature is not a claim that one vector “is” experience. It is a compact numerical proxy for the modeled trajectory: sensed content, expected content, error, memory change, attention, surprise, and learning considered together.

The important caution is that a memory embedding is not, by itself, an episode effect. The observed memory M_t is the memory of a particular subject after the episode has acted upon a prior state. It therefore contains at least two ingredients: what the subject brought into the run, and what the presented sequence changed. If two subjects end with different memories, that difference may be due to the episode, but it may also be due to different priors, vocabularies, sensor histories, or initial memory states.

For this reason we distinguish observed memory from state effect. A state-effect measure asks what changes when this episode is presented, relative to a matched baseline. In the simplest case we write

$$\Delta M_t = M_t^{\text{run}} - M_t^{\text{base}},$$

where the baseline preserves the relevant subject profile and prior state while removing, neutralizing, or otherwise controlling the presented sequence. The comparison then targets the contribution of the episode rather than the total content of the subject’s memory.

This distinction is especially important when comparing subjects. Raw memory answers: where is this subject now? State effect answers: what did this run do to this subject? The latter is usually the cleaner quantity when we want to compare the modeled consequences of an exposure across different profiles.

17 XVII. Reference Checks

The status of the reference report is easy to underestimate. It is not an illustration placed after the theory in order to make the theory feel concrete. It is one of the places where the theory is forced to take a definite form. A verbal claim can remain suggestive while leaving several implementations possible; a reference check must decide what counts, in a given run, as the relevant state, dependency, ordering, or absence.

We therefore treat these reports as small experimental benches for the framework. Each one fixes a minimal episode, a subject profile, and a quantity to inspect. The point is not to show that the system can draw a convincing picture, but to show that the model preserves the relation it says it preserves. If a report changes when only a renderer changes, the architecture

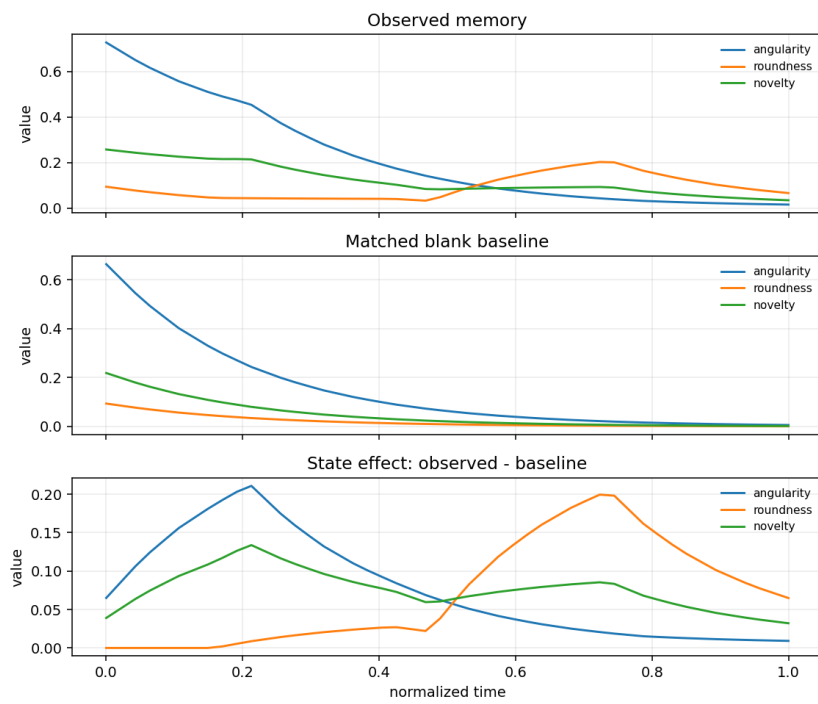


Figure 8: State effect subtracts a matched baseline from observed memory to isolate what the episode changed, rather than confusing episode consequence with prior state.

has been violated. If a claimed dependency cannot be made to appear under controlled conditions, the claim has not yet been operationalized.

This gives the reference section a methodological role. It converts philosophical commitments into inspectable failures. The reports are allowed to be mundane because their force is not aesthetic. Their force is that they make the implementation answerable to the formal story.

The invariants we check are deliberately low-level. First, expectation must be computed from the state carried into the moment, not from a memory already contaminated by the present input. In a repeated sequence, this is what lets the model distinguish genuine familiarity from a bookkeeping artifact. Second, expected input is not allowed to appear costlessly: it must pass through the internal expectation allocation, so reducing that allocation weakens the comparison rather than merely changing a display. Third, a view must be observationally inert. Re-rendering the same episode as a wall, a timeline, or a diagnostic panel may change pixels, but it may not change S_t .

We also check absence and collapse cases, because these are where false models often hide. If the subject has no sensor for a modality, the object remains present in the episode but has no subject-side sensory effect. If attention is set to zero, input opacity, encoding, learning impact, and topology deposit should collapse accordingly. Finally, changing priors should alter what memory and expectation bring to the encounter without changing the input effect attributable to the presented sequence itself. These are small tests, but they protect the central distinction: world sequence, subject profile, and rendered view are not interchangeable objects.

We can therefore read the reference suite as a set of causal probes. In an expectation-violation probe, we hold the sequence nearly constant and introduce a late mismatch; the relevant question is whether error rises at the violation rather than at mere repetition. In an unseen-modality probe, we present an object outside the subject’s sensorium; the object belongs to the episode, but should not enter the subject-side sensory state. In an attention-bottleneck probe, we reduce available capacity and ask whether sensing, encoding, learning, and deposit weaken together.

Other probes isolate less obvious commitments. An importance-weighted event should affect memory and learning in proportion to its authored weight only after it has been sensed and gated; importance is not a bypass around subject dependence. A valence-separation probe pairs predictability and desirability in crossed cases, showing that surprise and value are different coordinates of S_t . An objective-attention-shift probe lets the subject’s task reallocate attention, so that the same world sequence produces a different trajectory because a different relation to it is being enacted. Finally, a workspace-compression probe tests whether limited representation changes what is carried forward without pretending that the external object was simpler than it was. Together, these probes ask whether the framework’s

causal vocabulary is actually implemented.

18 XVIII. Source-Neutral Producers

We begin this section with a modest but important architectural point. The Episode contract is source-neutral not because all sources are identical, but because each source can expose a comparable subject-side trace. A native CaveNet run, a language-model run, and a conversation run may have different internal machinery, different temporal units, and different native state variables. The contract asks only that each producer adapt those internal representations into the common observation shape: time, expected input, actual input, attention, memory state, surprise, learning rate, active inputs, and metadata.

This adapter boundary is what lets us compare trajectories without pretending that a token, a spoken turn, and an authored Cave object have the same ontology. They need not. What must be shared is the episode surface through which the run becomes inspectable. In this sense, a producer is not a renderer and not a second model of experience. It is a translator from a source-specific process into the serialized trajectory format.

We therefore treat GPT-2, conversation, and CaveNet producers as different routes into the same downstream analysis space. Their meanings remain source-specific; their observations become structurally comparable.

For a GPT-2 producer, the token position can serve as the episode time index, while the presented token or its embedding supplies the actual input. The model’s teacher-forced next-token distribution gives us a principled expected input: the prediction is computed before the observed next token is incorporated. Hidden states can then be exposed as the producer’s memory state, attention matrices as source-native attention evidence, and token log-probability or embedding discrepancy as a surprise or prediction-error proxy. We should be careful here: transformer attention is not automatically identical to Cave attention as a subject-side gate. It is an adapted source variable, recorded with metadata so that later analysis can distinguish native mechanism from framework role.

A conversation producer performs a parallel translation at a different grain. Its time units are turns, utterances, or message segments rather than tokens. The actual input is the current turn, represented through lexical, semantic, speaker, or pragmatic features. Prior turns provide the context from which expectations are formed: anticipated topic, likely response type, emotional valence, or discourse continuation. Memory state may summarize retained conversational commitments, entities, goals, and unresolved tensions. In both cases, the producer does not claim that tokens and turns are ontologically equivalent. It claims only that each can be serialized as a subject-side trajectory.

Thus, source neutrality should be understood as a discipline of comparison rather than an erasure of difference. We do not ask a transformer, a dialogue system, and a CaveNet subject to possess the same inner ontology. We ask each to report, as faithfully as its substrate permits, the variables by which a temporal path can be inspected: what was presented, what was anticipated, what was retained, what was missed, and what changed as a consequence. The shared Episode surface gives us a common grammar for asking those questions.

This also places a burden on the adapter. It must not smuggle in equivalences that the source cannot justify. If a variable is a proxy, it should be marked as a proxy. If attention is architectural rather than functional, that distinction should remain visible. If time is token-based in one producer and turn-based in another, comparison must proceed with that difference in view. Source-neutrality therefore means that downstream tools can align observations without flattening the processes that produced them.

In this restricted sense, we compare like-shaped traces, not like-made mechanisms. The Episode contract supplies the shape; the producer meta-data preserves the mechanism. That separation is what allows comparison to become rigorous rather than merely visual.

19 XIX. Scope And Non-Claims

We should be explicit at the outset of these limits. Cave does not claim to detect consciousness in an external system, and it does not claim to manufacture consciousness by executing its update rules. Its object is more modest and, for that reason, more inspectable: a subject-side trajectory generated from an external sequence, a configured subject profile, and a set of state updates. This trajectory may be called experience-like only in the technical sense that it preserves relations normally hidden in a bare input log—sensing, attention, expectation, error, memory, value, and altered future coupling.

The distinction matters. A system can instantiate the Cave contract while remaining only a model of how appearances are selected and carried forward. Conversely, a conscious organism, if described in Cave terms, would not thereby have its consciousness explained away. The framework gives us a disciplined way to ask what was available to a subject, what was predicted, what was missed, what changed, and what consequences followed. It does not decide whether there is something it is like to undergo that trajectory.

Thus our claim is methodological rather than metaphysical. Before as-signing consciousness, denying it, or comparing it across systems, we first require an explicit object of comparison.

This requirement also fixes several negative commitments. We do not treat a feature vector as a natural unit of mentality. It is an encoding chosen for a model, useful because it lets us compute similarity, prediction, error,

and change. Another encoding might support the same explanatory role. Likewise, when we project a trajectory into a surface, manifold, corridor, wall, or other spatial figure, we are not claiming that the subject literally inhabits that geometry. The projection is an analytic instrument, not an inner room.

The same caution applies to generative systems used to produce, adapt, or describe episodes. If a language model supplies a sequence, labels features, or helps author an input stream, that production does not imply that the model underwent the resulting episode. It may generate a description of fear without fearing, a report of surprise without being surprised, or a sequence of memories without retaining them as its own.

Nor does an attractive visualization strengthen the metaphysical claim. A rendered trajectory can be illuminating, misleading, elegant, or opaque; none of those properties is evidence of consciousness. The evidence, such as it is, concerns whether the specified update relations were actually computed and serialized.

The positive claim, then, is deliberately narrow. We can model a subjective trajectory as the result of passing an external sequence through a configured subject: its sensors, attentional gates, workspace constraints, memory state, expectations, error computations, learning rule, value assignments, exposure policy, and topology-like accumulation. Each of these terms names a relation that can be specified without deciding whether the resulting process is conscious. The important point is not that the model discovers an inner essence, but that it prevents us from confusing the presented sequence with the sequence as available to a particular subject.

This is why the framework begins with transformations rather than declarations. An object may be present in the episode and absent from the subject if no sensor transduces it. A stimulus may be sensed but weakly encoded if attention is low. A familiar input may produce little error; a violated expectation may reorganize memory; a valued or aversive input may change future coupling. None of these consequences is guaranteed by the external stream alone. They arise from the stream as filtered, compared, weighted, retained, and carried forward by the configured subject. That transformed path is the claim's proper object.

20 XX. Conclusion

The central lesson of Cave is that Plato's image can be made operational. Instead of treating the cave as a metaphor to be interpreted from outside, we treat it as a specification for a run: an external episode, a configured subject, and a sequence of state updates that determine what is taken in, what is missed, what is predicted, and what is retained. This shift is deliberately modest. It does not solve consciousness by definition, and it does

not claim that a rendered scene is an experience. Rather, it identifies the computational relation that must be made explicit before such claims can be responsibly compared.

Our emphasis, therefore, has been architectural as much as philosophical. The model state is primary; every wall, dashboard, memory corridor, embedding, or topology is a projection from that state. Once the state-update chain is explicit, the old problem of appearances becomes an inspectable problem of sensors, attention, prediction, error, memory, value, and exposure. The cave is no longer only a place where shadows are seen. It becomes a controlled setting in which different subjects can traverse the same episode differently, and those differences can be serialized, tested, and compared.

At the end of this construction, the important object is neither the external sequence alone nor any single image by which it is displayed. It is the temporally extended relation between a world that offers events and a subject that can meet those events only through particular capacities. The subject has sensors rather than omniscience, attention rather than total uptake, expectations rather than neutral reception, memory rather than an empty slate, and values rather than indifferent registration. Each timestep therefore asks a specific question: given this subject's current state, what can become available, how strongly can it enter, what was anticipated, what was violated, what is learned, and how may future coupling be changed?

This is why we have treated the trajectory as the central computational object. It is not a hidden picture behind the rendered wall. It is a structured history of transformations: external availability $\rightarrow$ subject-side uptake $\rightarrow$ comparison with prior expectation $\rightarrow$ error, learning, retention, valuation, and altered exposure. The same episode can accordingly support multiple histories without contradiction. Difference is not added later by visualization; it is produced by the modeled relation itself.

The contribution, then, is a working formalism for asking how an external sequence becomes an internal history. It gives us a way to distinguish availability from uptake, uptake from expectation, expectation from error, and error from later consequence. In doing so, it replaces the idea of experience as a passive copy of input with the idea of a structured path: a temporally ordered, subject-dependent record of what could be sensed, what was attended, what was predicted, what mattered, and what changed.

This is a deliberately useful object. It can be serialized without a renderer, inspected without reducing it to a picture, and compared without pretending that all subjects received the same world in the same way. The formalism therefore makes room for both rigor and difference. It lets us say, with computational specificity, that two subjects shared an episode while not sharing the same trajectory through it.

If Cave succeeds, it does so by making the relation visible before making stronger claims about mind. It offers not a final theory of consciousness, but a disciplined substrate on which such theories can be stated, tested, and

compared. The modeled subjective trajectory is the first object we must get right.